$$\mathcal{C}(\tau) = \sum_i g_i^2 \frac{1}{Z_i} \mathrm{Tr}_{\mathcal{E}} \left\{ \mathrm{e}^{-\hat{H}_{\mathcal{E}}^{(i)}/k_{\mathrm{B}}T} \hat{\sigma}_z^{(i)}(\tau) \hat{\sigma}_z^{(i)} \right\}$$
$$= \sum_i g_i^2 \left\{ \cos\left(\Delta_i \tau\right) - \mathrm{i} \tanh\left(\frac{\Delta_i}{2k_{\mathrm{B}}T} \right) \sin\left(\Delta_i \tau\right) \right\}. \qquad (5.184)$$